

Report: ETL Traffic Data Analysis

Include your visualizations, analysis, results, insights, and outcomes. Explain your methodology and approach to the tasks. Add your conclusions to the sections.

1. Data Preparation

1.1. Loading the dataset

1.1.1. Loading Multiple CSV Files

Objective

The objective of this step was to load the California traffic collision dataset into PySpark DataFrames and prepare the data for further analysis. The assignment contains multiple CSV files representing different aspects of traffic collision information. Since the analysis is performed using PySpark, the datasets were loaded into PySpark DataFrames for efficient processing and analysis.

Methodology

- Created a Spark session for processing the datasets.
- Identified the available CSV files in the dataset folder.
- Loaded the CSV files using PySpark with headers enabled.
- Used schema inference to identify the appropriate data types.
- Created separate PySpark DataFrames for the different datasets.
- Examined the schema and sample records to verify successful data loading.

Observation

The datasets were successfully loaded into PySpark DataFrames. The schema and sample records confirmed that the required columns and data types were available for further data preparation and exploratory analysis.

Conclusion

Loading the datasets into PySpark provided a suitable foundation for processing the traffic collision data. The successful creation of the DataFrames confirmed that the PySpark environment and data-loading process were working correctly.

1.1.2. Dataset Schema Examination

Objective

The objective of this step was to examine the schema of the loaded California traffic collision datasets and understand the structure, columns, and data types available in each dataset. This verification was performed before proceeding with data cleaning and exploratory analysis.

Methodology

- Examined the schema of the Collisions dataset using `printSchema()`.
- Examined the schema of the Parties dataset.
- Examined the schema of the Victims dataset.
- Examined the schema of the Case IDs dataset.
- Reviewed the column names and their corresponding data types.
- Identified important fields related to collision severity, weather, road conditions, lighting, victims, parties, location, date and time.
- Verified that the datasets contained the required fields for subsequent analysis.

Observation

The schema examination showed that the datasets contain a combination of numerical, string, date and timestamp fields.

The Collisions dataset contains important attributes such as `case_id`, `county_location`, `weather_1`, `collision_severity`, `primary_collision_factor`, `road_surface`, `road_condition_1`, `lighting`, `latitude`, `longitude`, `collision_date` and `collision_time`.

The Parties dataset contains information such as `case_id`, `party_number`, `party_type`, `party_age`, `party_sex`, `party_sobriety`, vehicle-related information and party involvement details.

The Victims dataset contains information related to victims, including `case_id`, `party_number`, `victim_role`, `victim_sex`, `victim_age`, injury-related fields and safety-equipment information.

The Case IDs dataset contains the case identifier and associated date information.

Conclusion

The schema examination provided a clear understanding of the structure and variables available across the four datasets. The presence of collision, party, victim and case-level information provides the required foundation for performing data cleaning and subsequent exploratory data analysis.

1.1.3. Display Schema and Sample Data

Objective

The objective of this step was to examine the structure of the loaded traffic collision datasets and verify that the columns, data types, and sample records were correctly loaded. This helped ensure that the datasets were properly structured and ready for further data preparation and analysis.

Methodology

- Displayed the schema of the loaded DataFrames.
- Examined the column names and their corresponding data types.
- Displayed sample records from the datasets.
- Verified that the required fields were available for further analysis.
- Used the schema and sample records to understand the structure and content of the traffic collision data.

Observation

The schema and sample records provided an overview of the structure of the traffic collision datasets. The columns contained information related to collisions, parties, victims, and locations, which would be used in the subsequent data cleaning and exploratory data analysis steps.

Conclusion

Examining the schema and sample records confirmed that the datasets were structured appropriately for further processing. This step provided an understanding of the available variables and helped establish the foundation for subsequent data cleaning and analysis.

2. Data CleaningData Cleaning

2.1. Missing Values

Missing values were checked in the Collisions, Parties, Victims, and Case IDs datasets using PySpark.

The number of null values was calculated for each column. No sparse columns were removed at this stage.

The missing values were handled using the fillna() function. Missing values in the Collisions, Parties, and Victims datasets were replaced with "Unknown", while missing values in Case IDs were replaced with 0.

The collision_date column was also converted into the proper date format using to_date().

Result: Missing values were successfully handled, and the datasets were prepared for further data cleaning and analysis.

2.2. Fixing ColumnsFixing Columns

Duplicate records were removed from the Collisions, Parties, Victims, and Case IDs datasets using the dropDuplicates() function. Outliers were then detected in the numerical columns killed_victims and injured_victims using the Interquartile Range (IQR) method. The IQR was calculated using the 25th and 75th percentiles, with the standard $1.5 \times \text{IQR}$ rule used to identify values outside the lower and upper bounds. The analysis identified 0 outliers in killed_victims and 33,313 outliers in injured_victims.

Note: The identified outliers were detected for further analysis; they were not removed in this step.

Result: Duplicate records were removed and potential outliers were identified to improve the quality and reliability of the collision data.

2.3. Outliers

Analysis Outliers identified in the injured_victims column were removed using the Interquartile Range (IQR) method. The first quartile (Q1) and third quartile (Q3) were calculated using approxQuantile(). The IQR was calculated as: $\text{IQR} = \text{Q3} - \text{Q1}$. The lower and upper bounds were then calculated using the standard $1.5 \times \text{IQR}$ rule. Records with injured_victims values outside these bounds were filtered out. After removing the outliers, 902,248 records remained in the Collisions dataset.

Result: Outliers in the injured_victims column were successfully removed, resulting in a cleaner dataset for further exploratory data analysis.

3. Exploratory Data AnalysisExploratory Data Analysis

3.1.1 Data Preparation

Categorical and Numerical Variables

The variables were classified into categorical and numerical types to prepare the dataset for Exploratory Data Analysis.

Categorical variables such as weather_1, lighting, and road_condition_1 were encoded using PySpark's **StringIndexer**. New indexed columns were created with the suffix _index, while invalid values were retained using handleInvalid="keep".

The columns were then reordered in sorted order to provide a consistent and organized structure for further analysis.

Finally, the cleaned dataset was displayed and checked for its overall structure before proceeding with the Exploratory Data Analysis.

Result: Categorical encoding and column reordering were completed successfully.

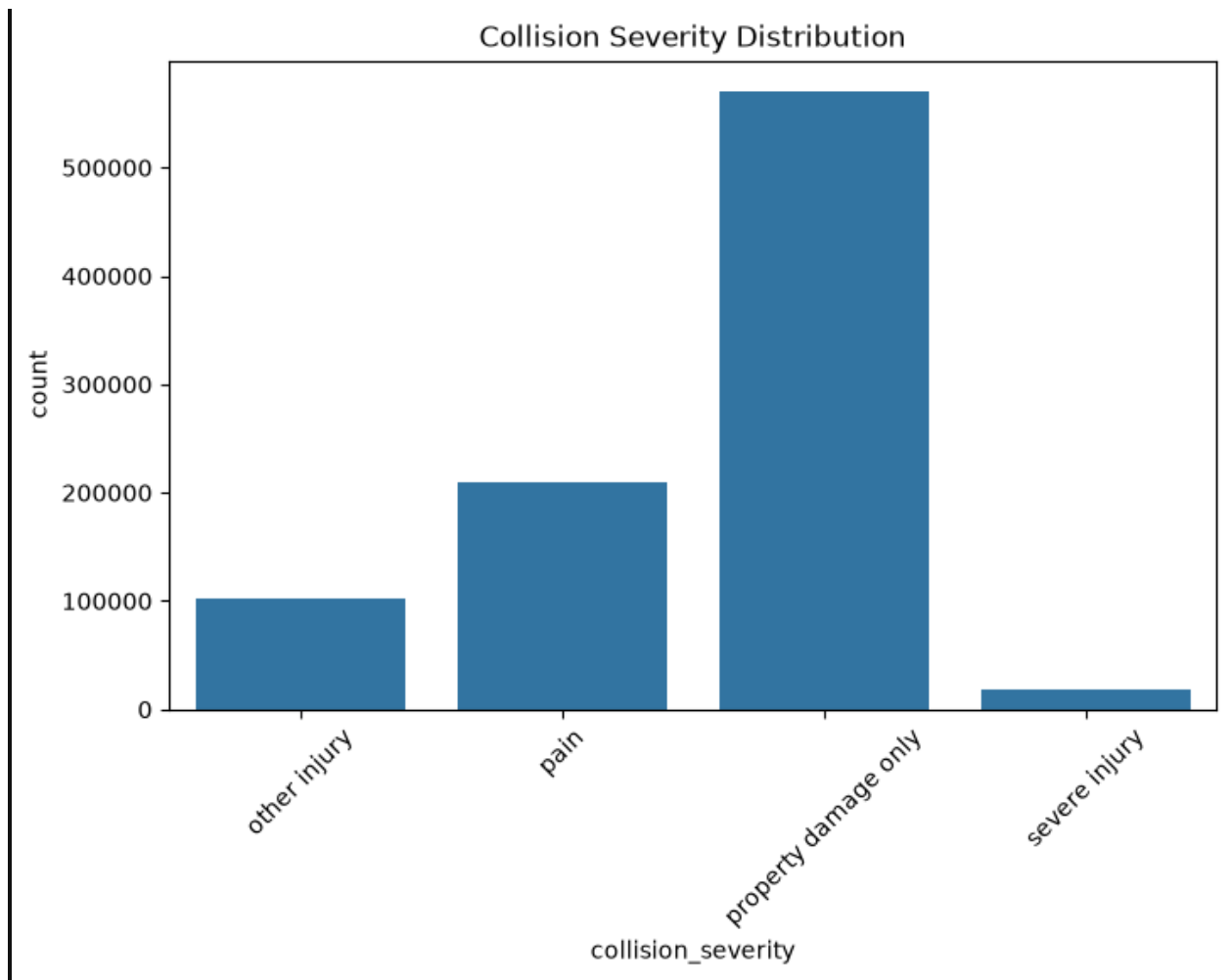
3.1.2 Analyze the Distribution of Collision Severity

The distribution of collision severity was analyzed using a bar chart based on the count of collisions in each severity category.

The analysis shows that **“property damage only”** is the most common collision severity category, followed by **“pain”** and **“other injury”**. **“Severe injury”** has the lowest number of collisions among the categories shown.

Overall, the distribution indicates that **most recorded collisions resulted in property damage only, while severe injuries were comparatively less frequent.**

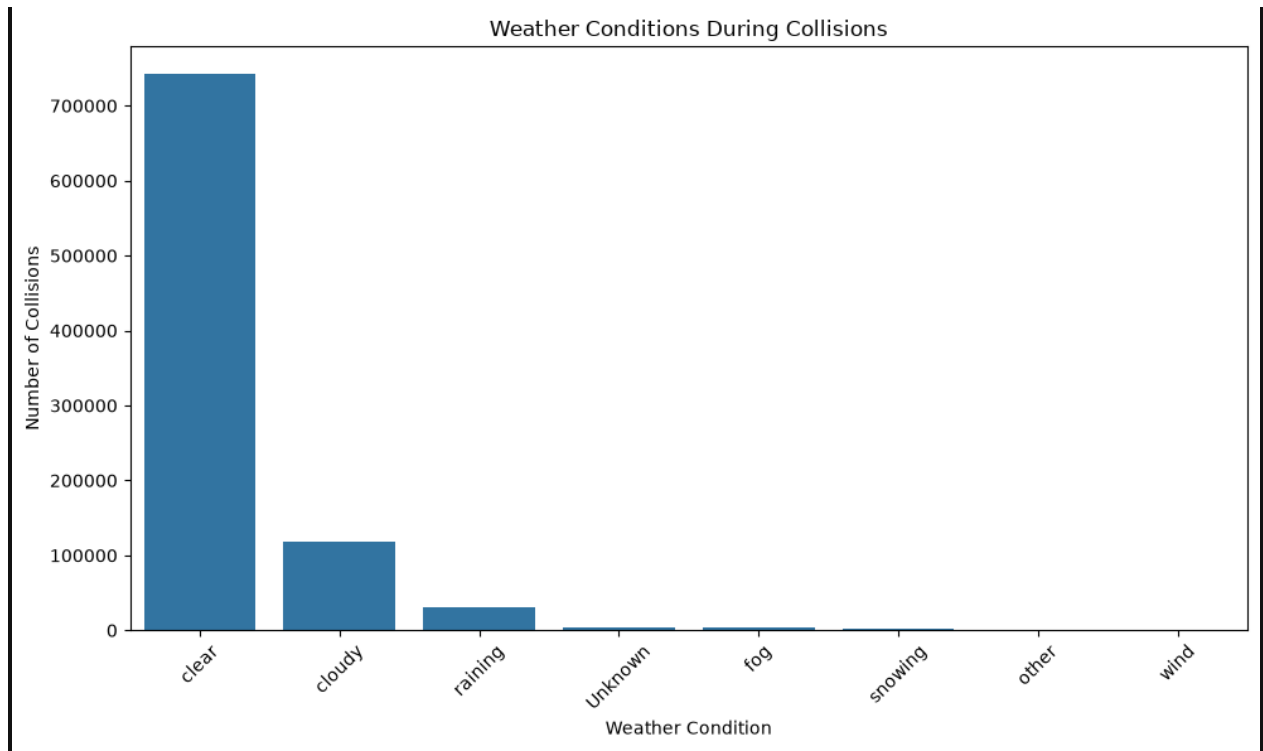
Figure: Collision Severity Distribution



3.1.3. Weather Conditions During

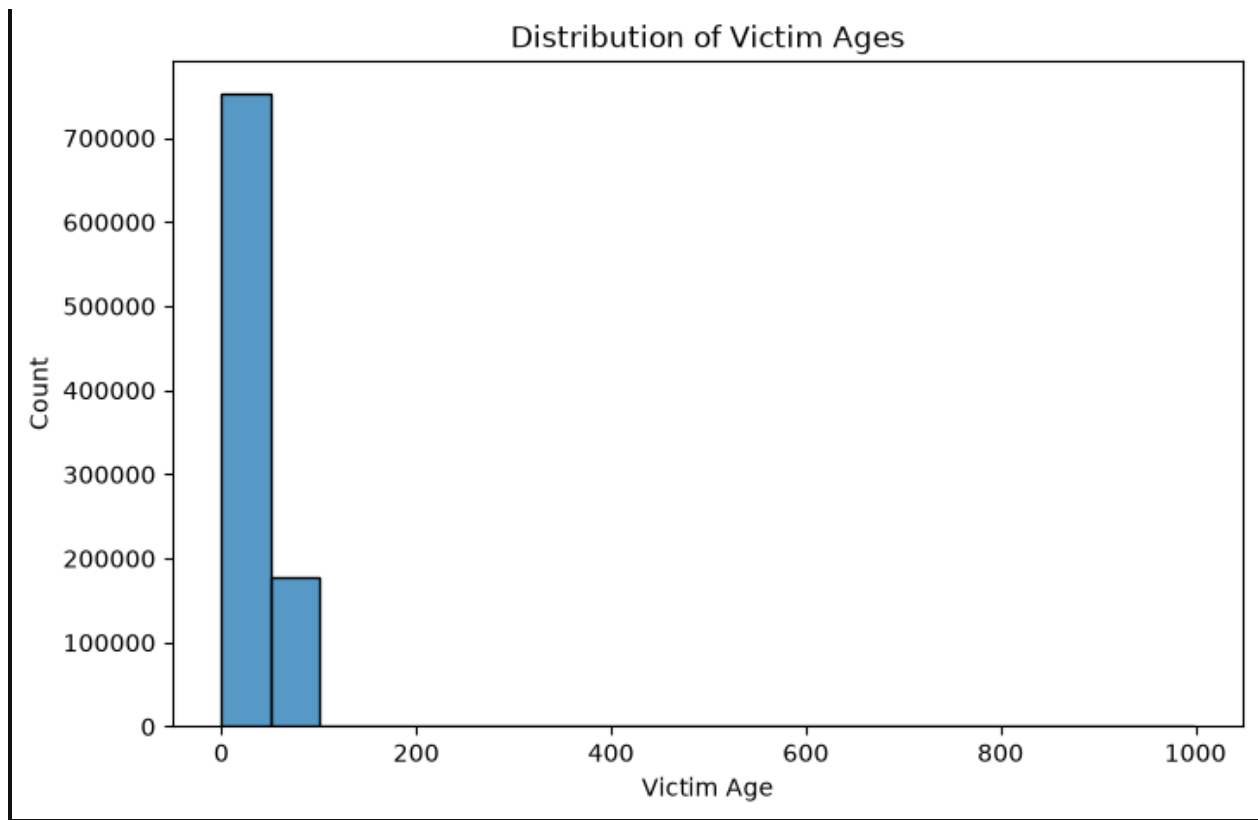
The weather conditions during traffic collisions were analyzed to understand the environmental conditions associated with collisions. The analysis shows that clear weather had the highest number of collisions (741,919), followed by cloudy weather (117,995) and rainy conditions (31,339). Other weather conditions such as unknown, fog, snowing, other, and wind had comparatively fewer collisions. Overall, most collisions in the dataset occurred during clear weather conditions.

Figure: Weather Conditions During Collisions



3.1.4. Victim Age Distribution

The distribution of victim ages was analyzed using a histogram. The graph shows that most victims are concentrated in the younger age groups, while the number of victims decreases as age increases. The distribution is highly concentrated at lower ages, with relatively few observations at higher ages.



3.1.5. Collision Severity vs Number of Victims

3.1.5. Collision Severity vs Number of Victims

Objective:

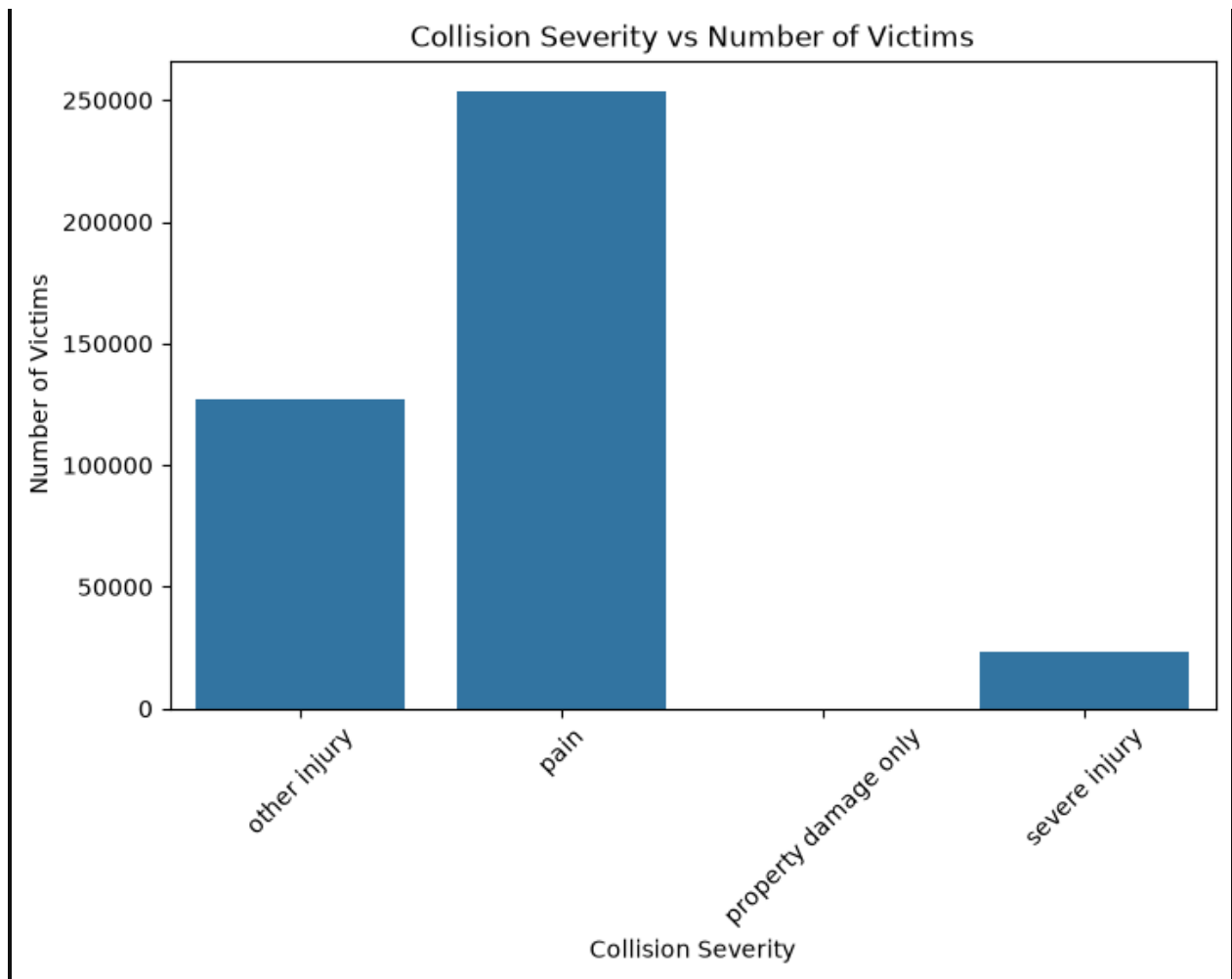
To study the relationship between collision severity and the number of victims involved in collisions.

Methodology:

The data was grouped by collision severity. Injured victims and killed victims were summed for each severity category, and a total victims column was calculated by adding injured and killed victims. A bar chart was used to compare the total number of victims across different collision severity categories.

Observation:

The analysis shows that collisions classified as “pain” have the highest number of victims, followed by “other injury”. “Severe injury” has a comparatively smaller number of victims, while “property damage only” has very few victims.



Conclusion:

The analysis indicates that collision severity is associated with the number of victims. Injury-related collision categories account for most of the victims, with "pain" having the highest total number of victims.

3.1.6. Weather Conditions vs Collision Severity

Objective:

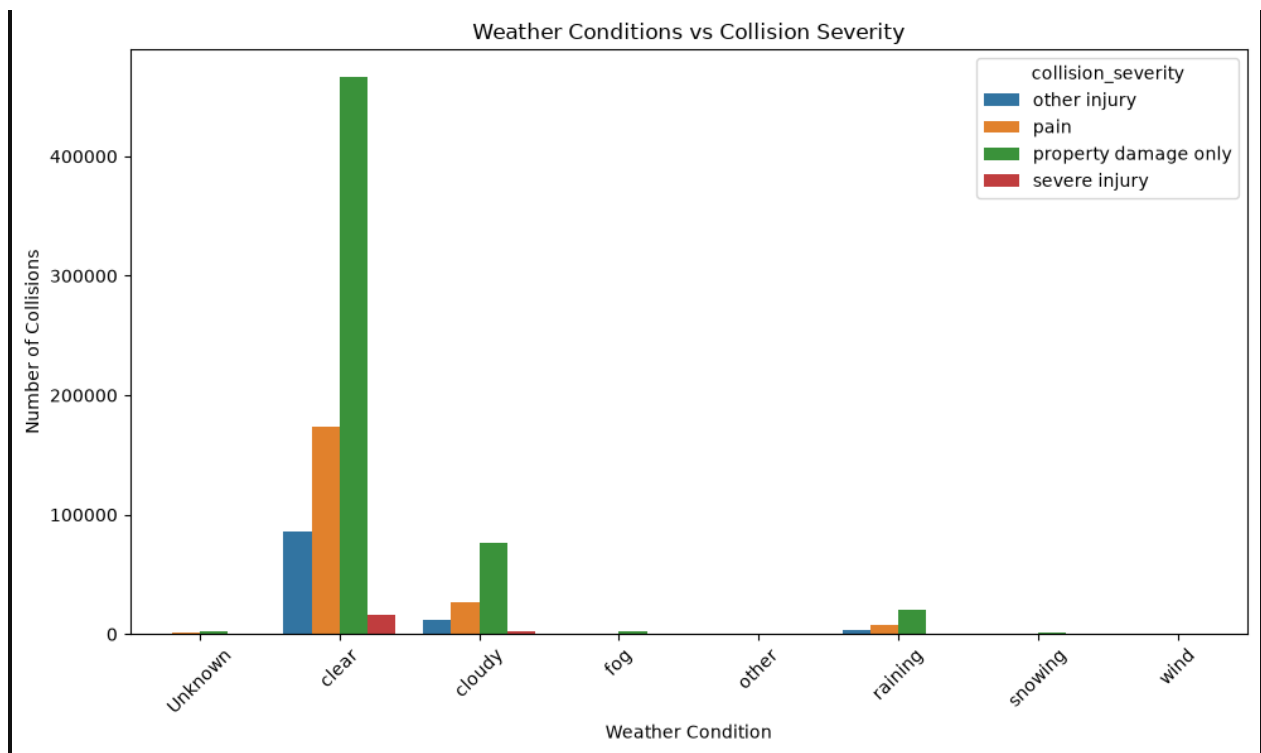
To analyze the relationship between weather conditions and collision severity.

Methodology:

The collision data was grouped by weather condition and collision severity. The number of collisions was counted for each combination. A grouped bar chart was used to compare collision severity across different weather conditions.

Observation:

The analysis shows that clear weather conditions have the highest number of collisions across the severity categories, followed by cloudy weather. Raining weather has comparatively fewer collisions, while fog, snowing, wind, and other conditions account for a much smaller number of collisions.



Conclusion:

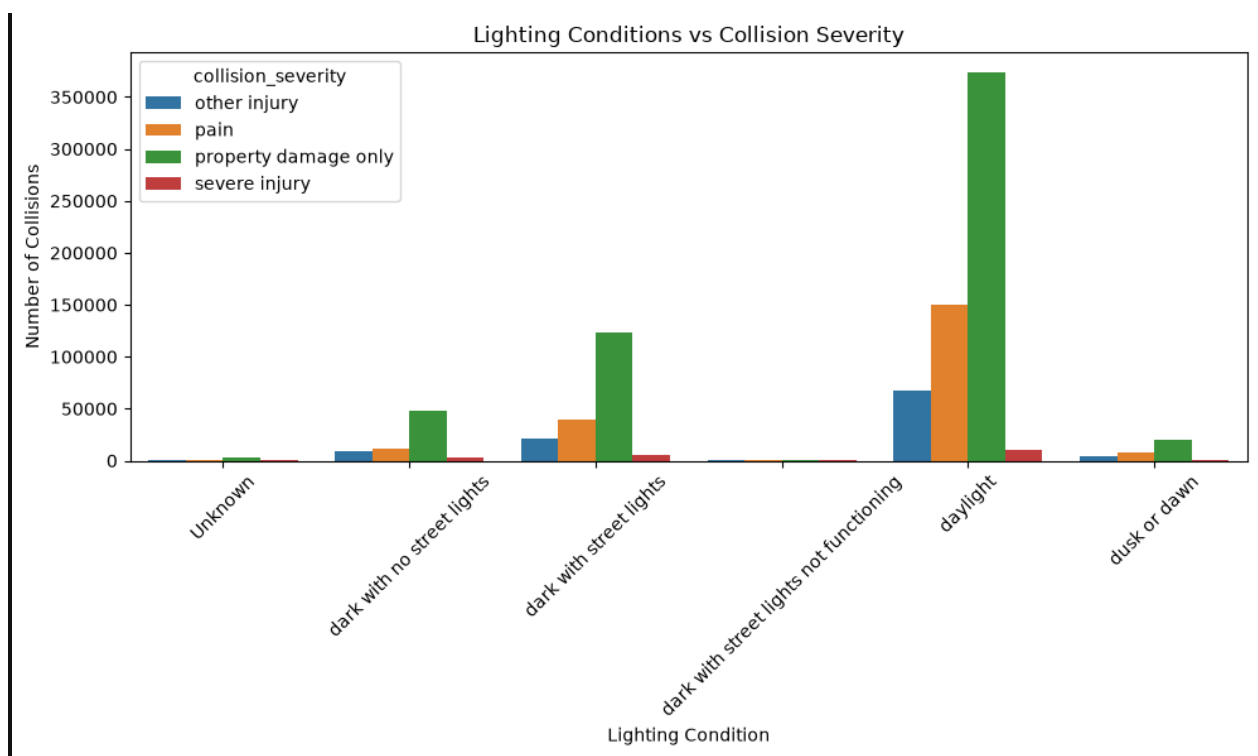
The analysis indicates that most collisions in the dataset occurred during clear and cloudy weather conditions. However, this does not necessarily mean that clear weather causes more severe collisions, as the number of collisions also depends on the overall exposure and frequency of each weather condition.

3.1.7. Lighting Conditions vs Collision Severity

Objective: To visualize the impact of lighting conditions on collision severity.

Methodology: The collision data was grouped by lighting condition and collision severity. The number of collisions was counted for each combination and visualized using a grouped bar chart.

Observation: The analysis shows that daylight has the highest number of collisions across the severity categories. Dark conditions with street lights also show a considerable number of collisions, while dusk or dawn and unknown lighting conditions have comparatively fewer collisions.



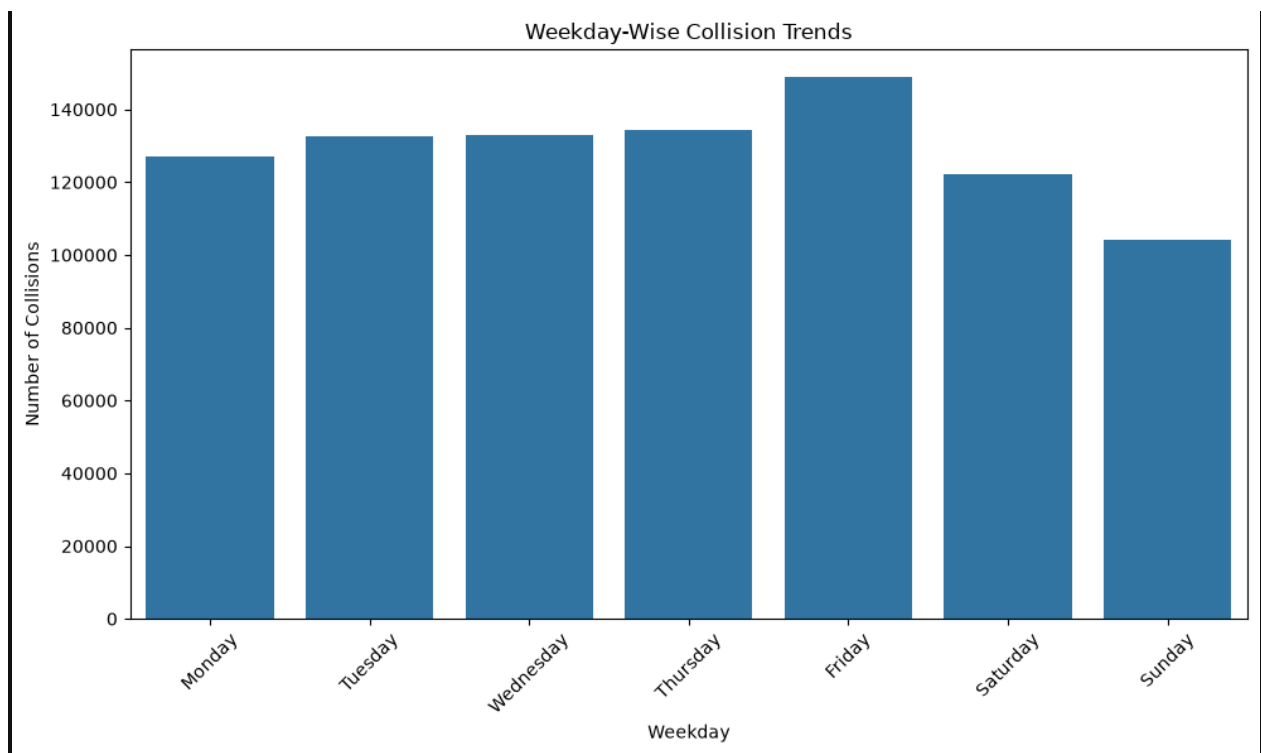
Conclusion: The results indicate that most recorded collisions occurred during daylight conditions. However, this does not necessarily mean that daylight increases collision severity, as traffic exposure and the frequency of each lighting condition can also affect the total number of collisions.

3.1.8. Weekday-Wise Collision Trends

Objective: To extract and analyze weekday-wise collision trends.

Methodology: The collision date was converted into the corresponding weekday. The collision records were then grouped by weekday and counted. The weekdays were arranged in the correct order from Monday to Sunday and visualized using a bar chart.

Observation: The analysis shows that Friday has the highest number of collisions, with approximately 149,000 collisions. Tuesday, Wednesday, and Thursday have similar collision counts of around 132,000–134,000. Monday has approximately 127,000 collisions, while Saturday has around 122,000. Sunday has the lowest number of collisions, with approximately 104,000.



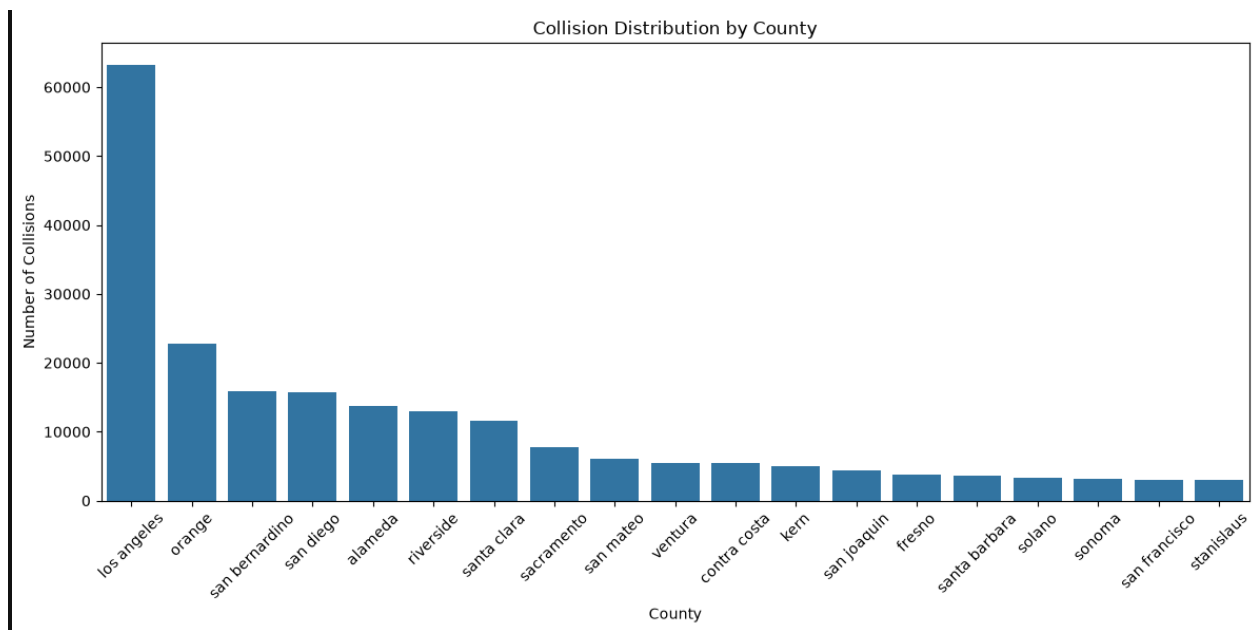
Conclusion: The results indicate that collision frequency is highest on Friday and lowest on Sunday. Overall, weekday collision counts are relatively similar from Monday to Thursday, followed by a noticeable increase on Friday and a decrease during the weekend.

3.1.9. Spatial Distribution of Collisions

Objective: To study the spatial distribution of collisions across different counties.

Methodology: The collision records were grouped by `caltrans_county` and the number of collisions was counted for each county. The results were sorted in descending order of collision count. The top 20 counties were selected and visualized using a bar chart.

Observation: The analysis shows that Los Angeles has the highest number of collisions among the counties shown, with approximately 63,235 collisions. Orange follows with 22,790 collisions, while San Bernardino and San Diego have approximately 15,943 and 15,701 collisions respectively. Alameda, Riverside, and Santa Clara also show relatively high collision counts. The remaining counties have progressively lower numbers of collisions.



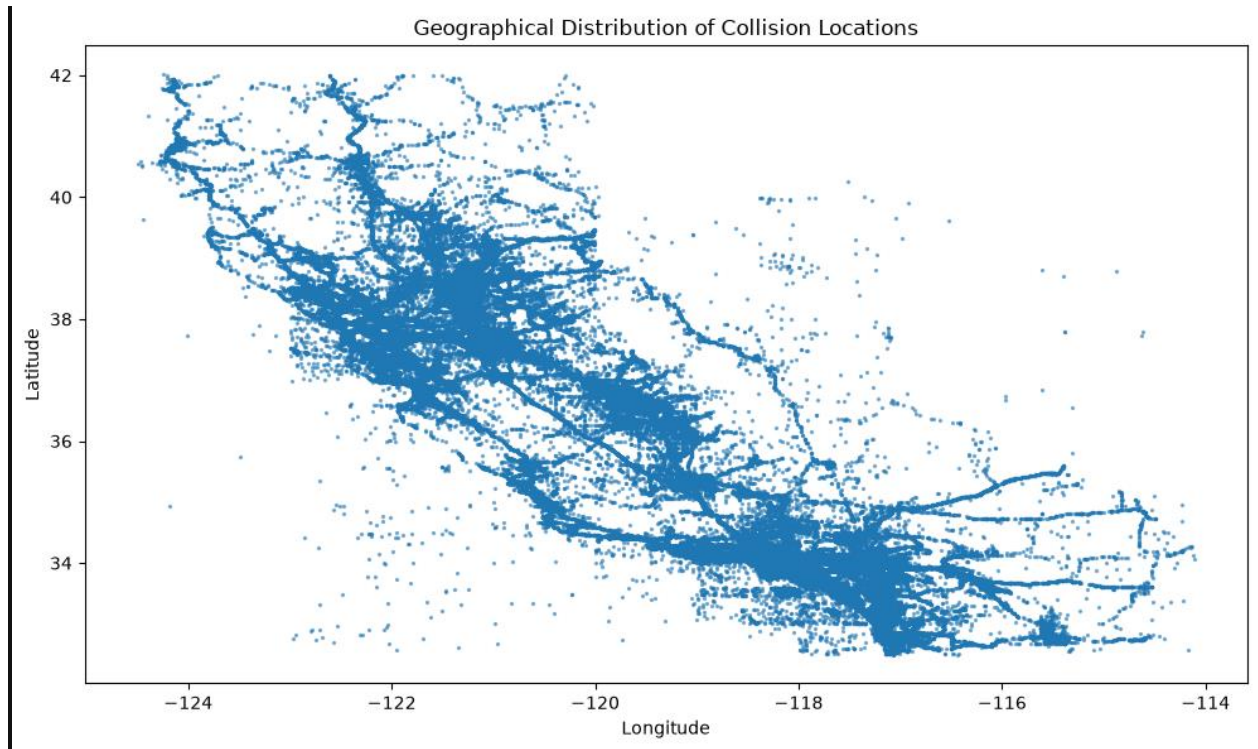
Conclusion: The collision distribution varies significantly across counties. Los Angeles has the highest number of reported collisions, followed by Orange and San Bernardino. This indicates that collisions are more concentrated in certain counties than others.

3.1.10. Collision Analysis by Geography

Objective: To analyze the geographical distribution of collision locations using latitude and longitude.

Methodology: The longitude and latitude columns were selected from the collision dataset and converted into numeric values. Invalid and missing coordinate values were removed. A scatter plot was then created using longitude on the x-axis and latitude on the y-axis to visualize the geographical distribution of collision locations.

Observation: The scatter plot shows that collision locations are distributed across a large geographical area of California. A high concentration of collision points can be observed in the central and southern parts of the state, while fewer collisions are visible in some northern and eastern areas.



Conclusion: The geographical analysis shows that collisions are not uniformly distributed across California. Certain areas have a much higher concentration of collision locations, indicating that collision incidents are more concentrated in specific regions.

3.1.11. Collision Trends Over Time

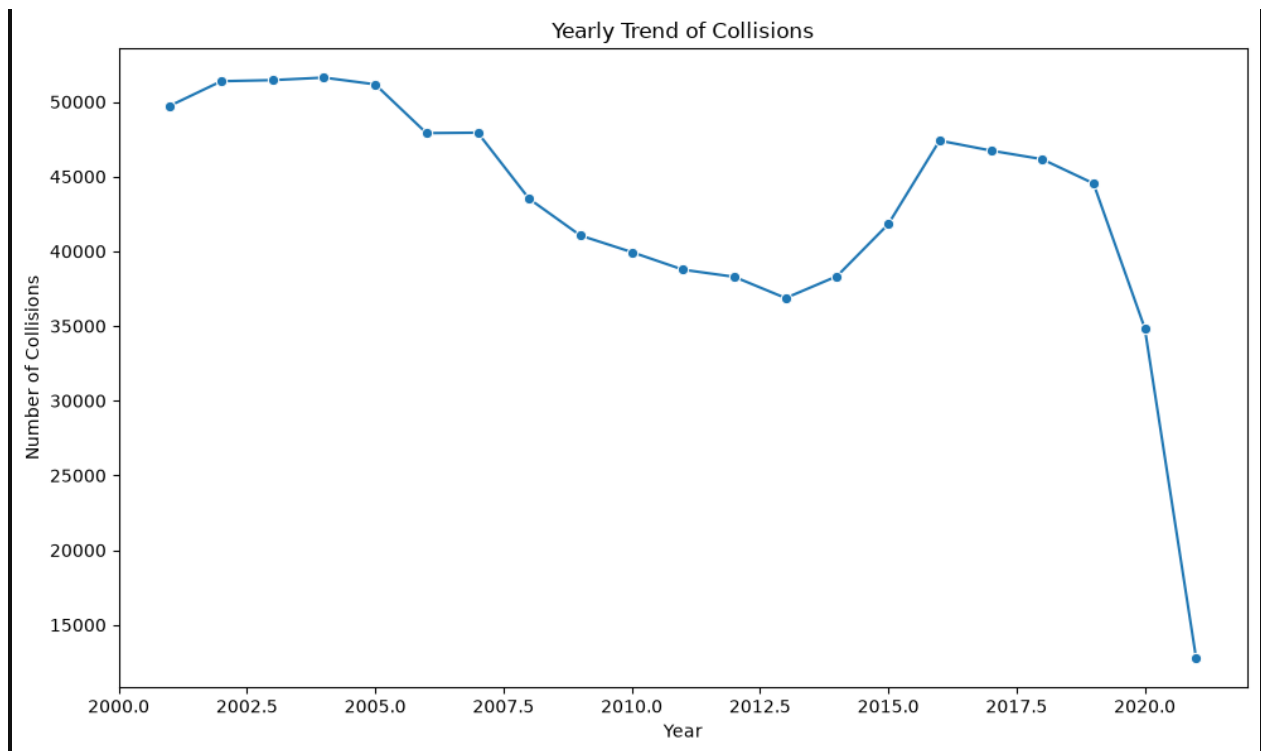
Objective: To extract and analyze traffic collision trends over time.

Analysis:

The collision date and collision time were converted into timestamp format. Year, month, and hour were extracted from the collision data. The data was then grouped by year, month, and hour to analyze the number of collisions during different time periods.

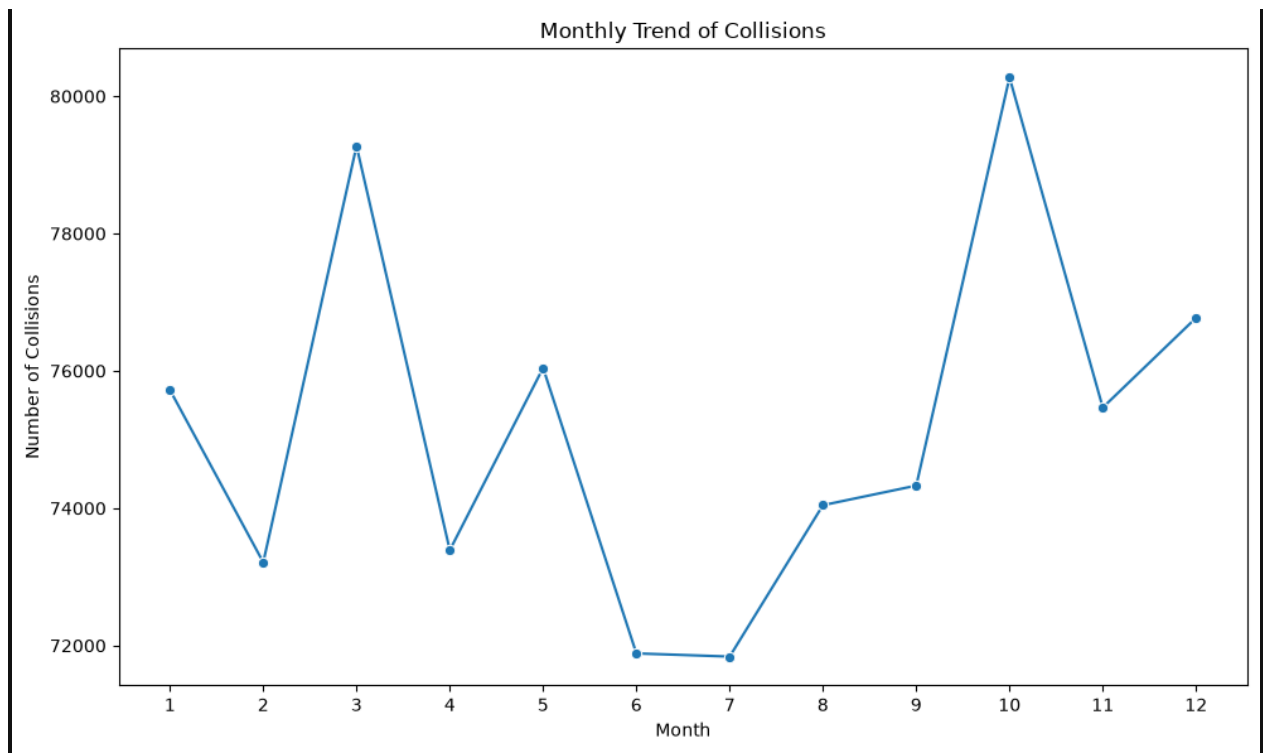
1. Yearly Trend:

The yearly trend shows that collisions were relatively high during the early 2000s, with the number of collisions reaching around 50,000 per year. A gradual decline can be observed after the mid-2000s, reaching a lower level around 2014. The number of collisions increased again after 2014 and reached approximately 45,000 around 2017. A sharp decline is visible in the final years of the dataset.



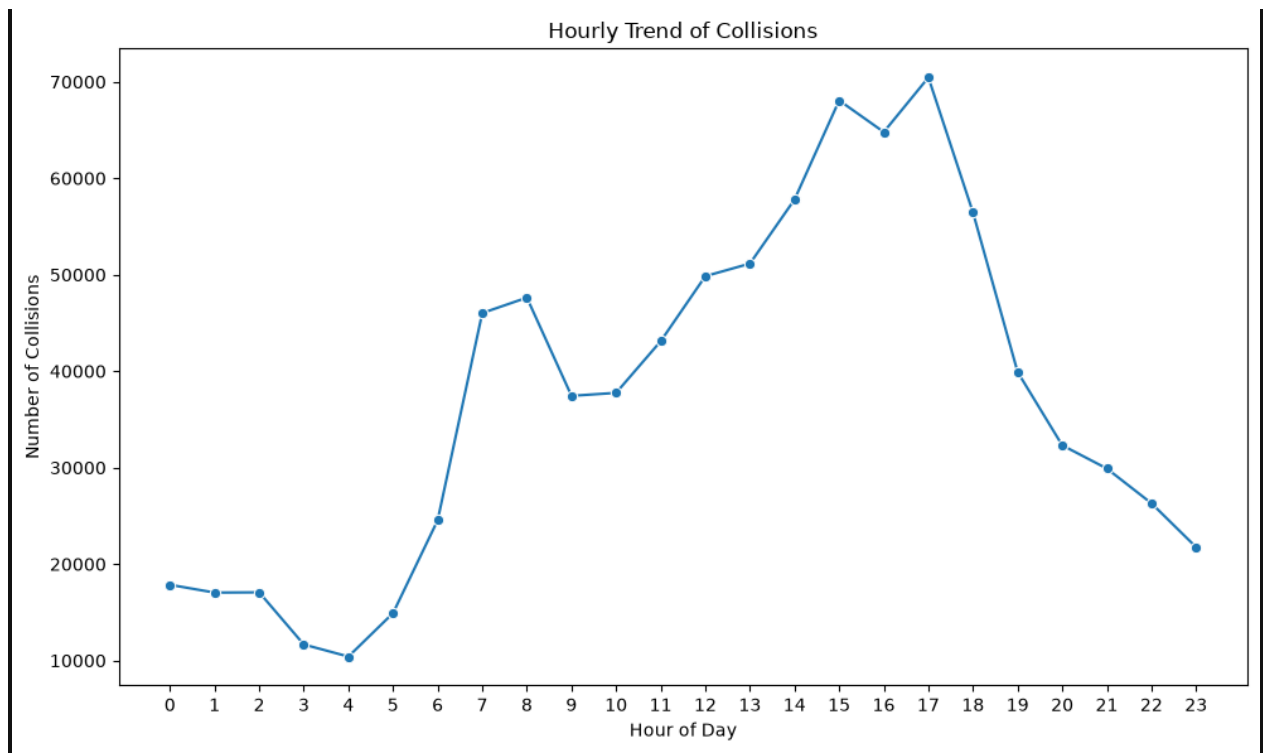
2. Monthly Trend:

The monthly trend shows variation in collision frequency across the 12 months. March has a relatively high number of collisions, while June and July show lower collision counts. October has the highest number of collisions, reaching approximately 80,000. November and December show a moderate increase compared with the previous months.



3. Hourly Trend:

The hourly trend shows a clear variation in collisions throughout the day. Collision counts are relatively low during the early morning hours, particularly around 3 AM–5 AM. The number of collisions increases significantly from the morning onwards. The highest collision count occurs around **5 PM (17:00)**, followed by a decline during the evening and night hours.



Conclusion:

The analysis shows that traffic collisions vary significantly by year, month, and hour. Collisions are generally more frequent during daytime and evening hours, with the highest hourly occurrence around 5 PM. Monthly patterns also show noticeable variations, with October having the highest collision count in the analyzed data. These temporal patterns can help identify high-risk periods and support traffic safety planning.

4.1. Top 5 Counties

Analysis:

The analysis identifies the counties with the highest number of recorded traffic collisions. Los Angeles County has the highest number of collisions with 63,235, followed by Orange County with 22,790, San Bernardino County with 15,943, San Diego County with 15,701, and Alameda County with 13,788. Records with a NULL county value are excluded from the Top 5 county ranking because they do not represent an identified county.

Conclusion:

Los Angeles County has the highest number of recorded traffic collisions among the identified counties, followed by Orange, San Bernardino, San Diego, and Alameda counties.

4.2. Month with Highest Collisions

Analysis:

The analysis identifies the month with the highest number of traffic collisions. The results show that **October (Month 10)** has the highest number of collisions, with **80,273 recorded collisions**.

Conclusion:

October has the highest number of traffic collisions in the dataset, indicating that it is the month with the highest collision frequency.

4.3. Weather Conditions with Highest Collisions

Analysis:

The analysis identifies the most common weather condition during traffic collisions. The results show that **clear weather** is the most common condition, with **741,901 recorded collisions**.

Conclusion:

Clear weather has the highest number of recorded traffic collisions in the dataset.

4.4. Fatal Collisions

Analysis:

The analysis was performed to calculate the percentage of traffic collisions that resulted in fatalities. The collision severity data contains categories such as Property Damage Only, Pain, Other Injury, and Severe Injury. No collision records were classified as "Fatal" in the dataset.

Therefore, the percentage of collisions resulting in fatalities is 0.0%.

Conclusion:

No fatal collisions were identified in the analyzed dataset.

4.5. Dangerous Time for Collisions

Objective:

To identify the most dangerous time of day based on the number of traffic collisions.

Analysis:

The collision data was grouped by hour and the total number of collisions was counted for each hour. The results were then sorted in descending order to identify the hour with the highest number of collisions.

The analysis shows that **17:00 (5:00 PM)** has the highest number of collisions, with a total of **70,448 collisions**.

Conclusion:

The most dangerous time of day for traffic collisions is **5:00 PM (17:00)**. This period records the highest number of collisions in the analyzed dataset, indicating that the evening period is a high-risk time for traffic collisions.

4.6. Road Surface Conditions

Objective:

To identify the top 5 road surface conditions with the highest collision frequency.

Analysis:

The collision data was grouped by road surface condition and the number of collisions was counted for each condition. The results were then sorted in descending order to identify the top 5 road surface conditions with the highest collision frequency.

The results are:

1. **Dry** – 814,906 collisions
2. **Wet** – 74,481 collisions
3. **NULL** – 7,848 collisions
4. **Snowy** – 3,991 collisions
5. **Slippery** – 1,013 collisions

Conclusion:

The analysis shows that **dry road conditions have the highest number of collisions (814,906)**, followed by wet road conditions with **74,481 collisions**. Snowy and slippery conditions have considerably fewer recorded collisions.

4.7. Lighting Conditions

Objective:

To analyze the lighting conditions that contribute to the highest number of traffic collisions.

Analysis:

The collision data was grouped by lighting condition and the number of collisions was counted for each condition. The results were sorted in descending order to identify the top 3 lighting conditions with the highest collision frequency.

The results are:

1. **Daylight** – 602,889 collisions
2. **Dark with street lighting** – 188,735 collisions
3. **Dark with no street lighting** – 72,223 collisions

Conclusion:

The analysis shows that **daylight conditions have the highest number of collisions**, with 602,889 recorded collisions. Dark conditions with street lighting have the second-highest number, followed by dark conditions with no street lighting.

5. Conclusion

Final Insights and Recommendations

The analysis of traffic collision data reveals several important patterns and trends across time, road conditions, and lighting conditions.

1. Yearly Trend:

2. The number of collisions varied significantly across the years. Collision counts were relatively high during the early years, followed by a gradual decline and fluctuations in later years.

3. Monthly Trend:

Collision frequency varied across the 12 months. October recorded the highest number of collisions, while June and July had relatively lower collision counts.

4. Hourly Trend:

The most dangerous time of day was **5:00 PM (17:00)**, with **70,448 collisions**. This indicates that the evening period is a high-risk time for traffic collisions.

5. Road Surface Conditions:

The highest collision frequency was observed on:

- a. **Dry:** 814,496
- b. **Wet:** 74,481
- c. **Unknown (NULL):** 7,848
- d. **Snowy:** 3,991
- e. **Slippery:** 1,013

Dry and wet road surfaces had the highest recorded collision frequencies.

6. Lighting Conditions:

The highest number of collisions occurred during:

- a. **Daylight:** 602,889
- b. **Dark with street lights:** 188,735
- c. **Dark with no street lights:** 72,223

Daylight conditions recorded the highest number of collisions in the analyzed data.

7. Fatal Collisions:

No collision records were classified as **Fatal** in the dataset. Therefore, the calculated percentage of fatal collisions was **0.0%**.

Recommendations

- Road safety measures should be strengthened during evening peak hours.
- Drivers should be more cautious during wet and slippery road conditions.
- Proper street lighting should be maintained in areas with frequent nighttime collisions.

- Traffic monitoring and road-safety awareness campaigns should focus on high-risk periods and locations.
- The identified collision patterns can help authorities develop better road-safety strategies and reduce traffic accidents.